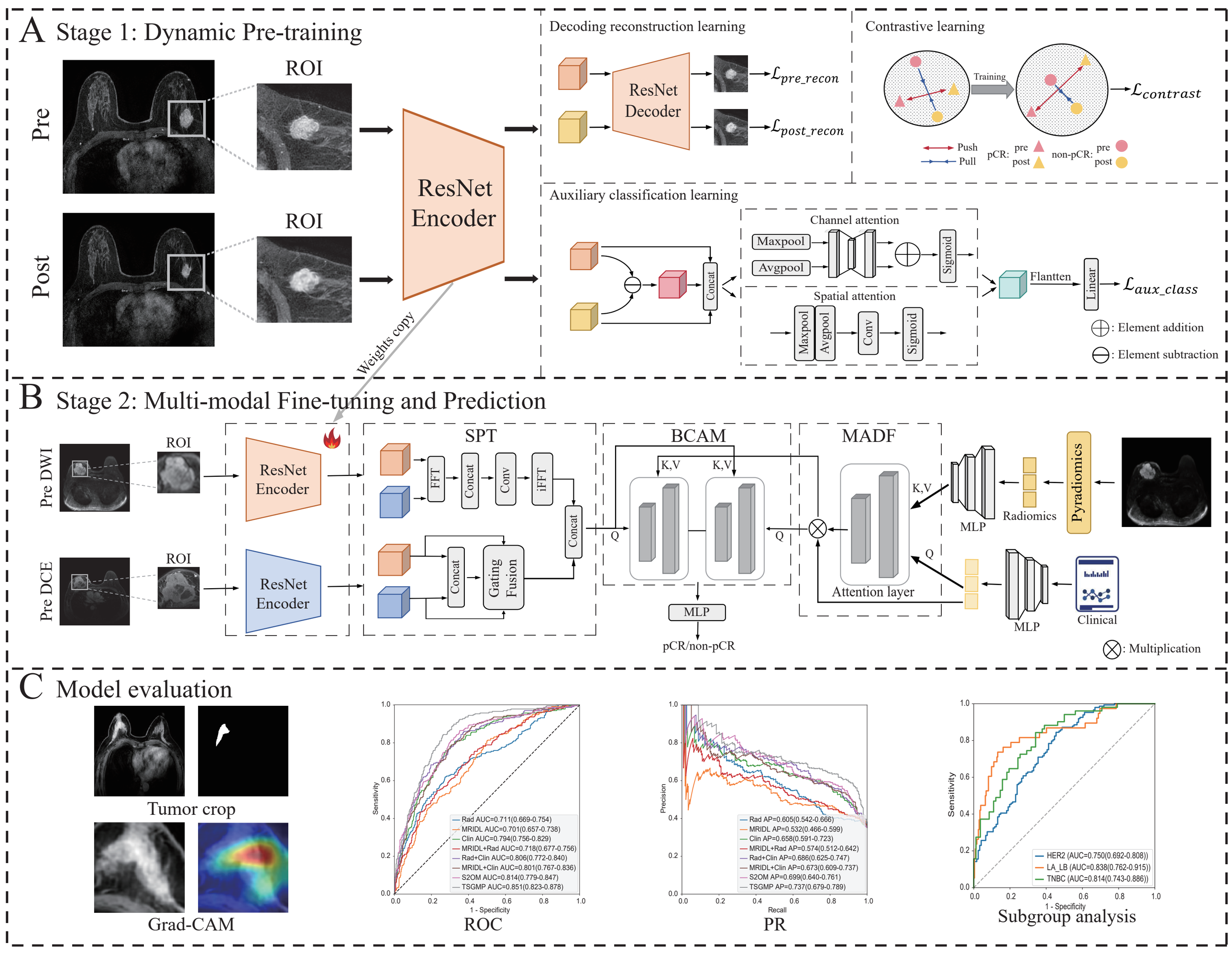
A Stage 1: Dynamic Pre-training
Pre
Post
ROI
ROI
ResNet Encoder
Weights copy
Decoding reconstruction learning
ResNet Decoder
$\mathcal{L}_{pre_recon}$
$\mathcal{L}_{post_recon}$
Contrastive learning
Training
$\mathcal{L}_{contrast}$
Push
Pull
pCR: pre post
non-pCR: pre post
Auxiliary classification learning
Concat
Channel attention
Maxpool
Avgpool
Sigmoid
Spatial attention
Maxpool
Avgpool
Conv
Sigmoid
Flantten
Linear
$\mathcal{L}_{aux_class}$
: Element addition
: Element subtraction
B Stage 2: Multi-modal Fine-tuning and Prediction
Pre DWI
Pre DCE
ROI
ROI
ResNet Encoder
ResNet Encoder
SPT
FFT
Concat
Conv
iFFT
Concat
Concat
Gating Fusion
BCAM
K,V
K,V
Q
Q
MLP
pCR/non-pCR
MADF
K,V
Q
Attention layer
MLP
Radiomics
Pyradiomics
MLP
Clinical
: Multiplication
C Model evaluation
Tumor crop
Grad-CAM
Sensitivity
1 - Specificity
Rad AUC=0.711(0.669-0.754)
MRIDL AUC=0.701(0.657-0.738)
Clin AUC=0.794(0.756-0.829)
MRIDL+Rad AUC=0.718(0.677-0.756)
Rad+Clin AUC=0.806(0.772-0.840)
MRIDL+Clin AUC=0.801(0.767-0.836)
S2OM AUC=0.814(0.779-0.847)
TSGMP AUC=0.851(0.823-0.878)
ROC
Precision
Recall
Rad AP=0.605(0.542-0.666)
MRIDL AP=0.532(0.466-0.599)
Clin AP=0.658(0.591-0.723)
MRIDL+Rad AP=0.574(0.512-0.642)
Rad+Clin AP=0.686(0.625-0.747)
MRIDL+Clin AP=0.673(0.609-0.737)
S2OM AP=0.699(0.640-0.761)
TSGMP AP=0.737(0.679-0.789)
PR
Sensitivity
1 - Specificity
HER2 (AUC=0.750(0.692-0.808))
LA_LB (AUC=0.838(0.762-0.915))
TNBC (AUC=0.814(0.743-0.886))
Subgroup analysis